

# Application of Integrals: Formula Sheet

## Class 12 Mathematics – Chapter 8 Complete Reference & Formulas

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### 1 Area Under Simple Curves

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- **1. Area Bounded by  $y = f(x)$ ,  $x$ -axis, and Ordinates  $x = a, x = b$ :** Using vertical elementary strips of height  $y$  and width  $dx$  ( $dA = y dx$ ):

$$\text{Area } A = \int_a^b dA = \int_a^b y dx = \int_a^b f(x) dx$$

- **2. Area Bounded by  $x = g(y)$ ,  $y$ -axis, and Abscissae  $y = c, y = d$ :** Using horizontal elementary strips of length  $x$  and width  $dy$  ( $dA = x dy$ ):

$$\text{Area } A = \int_c^d dA = \int_c^d x dy = \int_c^d g(y) dy$$

- **3. Curves Below the  $x$ -axis ( $f(x) < 0$ ):** Since area is a non-negative geometric quantity, take the absolute value:

$$\text{Area } A = \left| \int_a^b f(x) dx \right|$$

- **4. Curves with Portions Above and Below the  $x$ -axis:** If curve crosses the  $x$ -axis partitioning the region into  $A_1 < 0$  and  $A_2 > 0$ :

$$\text{Total Area } A = |A_1| + A_2 = \left| \int_a^c f(x) dx \right| + \int_c^b f(x) dx$$

## 2 Standard Geometrical Areas Evaluated by Integration

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- **1. Standard Circle ( $x^2 + y^2 = a^2$ ):** By symmetry about both coordinate axes:

$$\text{Total Area} = 4 \int_0^a y \, dx = 4 \int_0^a \sqrt{a^2 - x^2} \, dx = \pi a^2$$

- **2. Standard Ellipse ( $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ):** By symmetry in all four quadrants:

$$\text{Total Area} = 4 \int_0^a y \, dx = 4 \cdot \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx = \pi ab$$

- **3. Standard Parabola ( $y^2 = 4ax$ ):** Bounded by the curve,  $x$ -axis and line  $x = x_0$ :

$$\text{Area} = \int_0^{x_0} \sqrt{4ax} \, dx = 2\sqrt{a} \cdot \left[ \frac{2}{3} x^{3/2} \right]_0^{x_0} = \frac{4\sqrt{a}}{3} x_0^{3/2}$$

## 3 Integration Formulas Essential for Area Computations

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- **Standard Circular / Elliptic Arc Integral:**

$$\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left( \frac{x}{a} \right) + C$$

*Definite Form over Quadrant  $[0, a]$ :*

$$\int_0^a \sqrt{a^2 - x^2} \, dx = \left[ \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left( \frac{x}{a} \right) \right]_0^a = \frac{\pi a^2}{4}$$